TEXORA · LECTURE SHEET

Overall Project Plan for a Composite Textile Industry

Designing an integrated mill on paper — from the market backwards: product and capacity choice, the backward machine-balance calculation through dyeing, weaving and spinning, utilities and site, manpower, cost and implementation schedule.

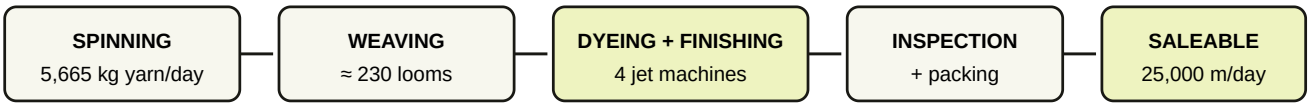
TE-4107 · PRODUCTION PLANNING AND CONTROL · TOPIC 9 · LEVEL 4 TERM 1

1. What "composite" means and how such a project is planned

A **composite textile industry** integrates successive stages under one roof and one management — typically spinning, weaving (or knitting), wet processing and finishing, sometimes garmenting. Integration buys control over quality and lead time at the cost of capital and complexity. An **overall project plan** (feasibility-level plan) must answer, in order: What will be sold, to whom, how much? What must each department produce to feed the next? What machines, utilities, people, land and money does that require? In what sequence is it built and commissioned? The planning direction is **backwards**: begin with the saleable product and work upstream, because every department exists to feed the one after it.

2. Step 1 — Product, market and finishing capacity

Fix the product mix first: here, a single representative product is used so the arithmetic stays visible — finished woven fabric, average 220 g/m, target output **25,000 m/day** (≈ 5,500 kg/day), 300 working days/year. Real plans carry several products; each is converted to a common load (kg/day or machine-hours) before balancing. Finishing/dyeing is the pacing department in most composite mills (batch machines, longest changeovers), so its capacity is fixed first and everything else is balanced to it.



Material flows left → right; the plan is calculated right → left, each stage grossed up for its losses.

Fig. 1 — The composite chain and its balancing direction.

3. Step 2 — The backward machine balance (worked)

WORKED CALCULATION — BALANCING THE WHOLE MILL TO 25,000 M/DAY

Finished fabric: 25,000 m/day × 0.220 kg/m = **5,500 kg/day**.

Dyeing/finishing: jet machines of 500 kg per batch, 3 batches/day → 1,500 kg per machine-day. Machines = 5,500 ÷ 1,500 = 3.67 → **4 machines** (one part-loaded machine is the department's cushion).

Weaving: yarn requirement = fabric × (1 + weaving and processing waste allowance 3 %) = 5,500 × 1.03 = **5,665 kg/day of yarn** (warp + weft). Loom output: 220 picks/min ÷ 2,400 picks per metre (24 picks/cm) = 0.0917 m/min = 5.5 m/h; over a 20-h effective day = 110 m/loom/day. Looms = 25,000 ÷ 110 = 227 → **230 looms** installed.

Spinning: average yarn 20 tex. Per spindle production = delivery 18 m/min × 60 × 20 g/km = 21.6 g/h; over a 22-h effective day ≈ 0.475 kg/spindle/day. Spindles = 5,665 ÷ 0.475 ≈ 11,920 → **12,000 spindles** (10 ring frames of 1,200), plus the matched blowroom–card–drawframe–comber line sized to feed them.

Every arrow in Fig. 1 is now a number, and the numbers interlock: change the fabric target and the whole chain re-derives.

Supporting departments are balanced the same way: sizing capacity to beam demand of 230 looms; inspection tables to 25,000 m/day; the boiler house to the dyeing department's steam profile (the TE-4106 boiler study is exactly this sizing exercise at small scale); the ETP to the dye-house effluent volume; humidification to shed air changes (TE-4106, Study B).

4. Steps 3–6 — Site, utilities, manpower, money, schedule

PLAN ELEMENT	CONTENTS	DERIVED FROM
Site and layout	Land, building areas by department, straight-line flow, expansion reserve; effluent plant at the wet end	Machine counts + material flow (Topic 1 layout principles)
Utilities	Steam (boiler capacity from dyeing load), power, water and its treatment, compressed air, humidification, ETP	Process loads per department, TE-4105/4106 methods

Manpower	Operators by machine assignment (interference-aware, Topic 5), technicians, quality, PPC and management; shift pattern	Machine counts × staffing norms; HOK used as the sanity check on total operator-hours
Project cost and finance	Land and building, machinery (the dominant share), utilities, erection, pre-production expenses, working capital; sources of finance and repayment	Vendor quotations + the balances above
Production cost and viability	Cost per metre from material (the largest share), conversion and overhead; selling price comparison; break-even utilisation of the new plant	Costing methods of TE-3109; break-even logic at plant scale
Implementation schedule	Phases: land and design → civil works → utility installation → machinery erection (in process order) → trial production → commercial production; critical path of the project itself	Delivery and erection lead times; utilities must precede the machines they serve

Project-planning pitfalls the plan must guard against: balancing every department at exactly 100 % of the pacing department (no cushion anywhere — the first breakdown starves the chain); sizing utilities to average rather than peak simultaneous load; forgetting working capital (a balanced mill with no money for cotton is a museum); and planning capacity for the launch product mix only, with no flexibility for the mix the market actually orders in year two.

Glossary

Composite industry — an enterprise integrating successive production stages under one management. **Machine balance** — the calculation matching each department's capacity to the next department's demand, allowing for losses. **Pacing department** — the department whose capacity sets the whole plant's rate. **Capacity cushion** — deliberate spare capacity absorbing breakdowns and mix changes. **Working capital** — money locked in raw materials, work-in-progress, finished stock and receivables. **Project plan** — the integrated document of product, capacities, site, utilities, manpower, cost and schedule for a new plant.

Key points

• Plan backwards from the saleable product; fix the pacing department (usually dyeing/finishing) first.
• Worked balance: 25,000 m/day → 5,500 kg/day finished → 4 jet dyeing machines; 5,665 kg/day yarn → 230 looms and ≈ 12,000 spindles.
• Utilities, manpower and money are all derived from the machine balance — never guessed separately.
• A plan balanced at exactly 100 % everywhere, or without working capital, is a plan to fail politely.

• Implementation follows process logic: utilities before machines, trial production before commercial promises.

Practice

1. Rebalance the dyeing department of the worked example for a target of 33,000 m/day of the same fabric: finished kg/day, and jet machines needed at the same batch pattern.
2. Why is the plan calculated backwards (finishing → spinning) when material flows forwards?
3. A planner sizes the boiler exactly to the dye house's average steam demand. Give two operational consequences.
4. Which single number in the worked example would you challenge first as a reviewer, and why?

Answer key

1. Finished fabric = 33,000 × 0.220 = 7,260 kg/day. Jet capacity per machine-day = 1,500 kg → 7,260 ÷ 1,500 = 4.84 → 5 machines. (And the rest of the chain re-derives with them: yarn = 7,260 × 1.03 ≈ 7,478 kg/day, looms ≈ 300, spindles ≈ 15,750 — a reviewer should expect the candidate to note that the whole balance moves, not just dyeing.)
2. Because demand originates at the saleable end: only the market fixes how much finished fabric must leave the gate. Each upstream department's required output is then the next department's input grossed up for that stage's losses. Calculating forwards from, say, a convenient number of spindles produces a plant whose finishing end is either starved or flooded — capacity that exists where it is not needed and bottlenecks where it is.
3. (i) Peak coincidence: dyeing machines call for steam in overlapping ramps (heating phases coincide after loading), so simultaneous demand exceeds the average — pressure drops, dyeing cycles stretch and shade reproducibility suffers exactly at the busiest moment. (ii) No margin for fouling, a tube leak or one boiler under maintenance: any derating stops the pacing department and therefore the whole mill. Boilers are sized to peak simultaneous load plus a maintenance margin (commonly n+1 thinking), with averages used only for fuel budgeting.
4. Strong answers challenge the 3 % combined waste allowance or the 20-h "effective day" for looms: both are single assumptions carrying the whole upstream balance — a 2-point error in waste moves yarn demand by about 110 kg/day (≈ 230 spindles); an effective day of 18 h instead of 20 raises loom need from 227 to about 253. A reviewer tests the assumptions with the largest leverage on machine counts, because those are what money is spent on. (Other defensible choices: the 500 kg × 3 batch dyeing pattern, or the 21.6 g/h spindle production, with similar leverage reasoning.)